

ENG. ALEJANDRO DE LA ROCHA

Full-stack Software Engineer | 11+ years of experience

AlejandroDeLaRocha.com | hi@alejandrolarocha.com | [YouTube Mentor/Tutorial Channel](#)

Professional Summary

Full-stack Software Engineer with 8+ years of experience in frontend and backend development across healthcare, SaaS, and enterprise systems.

Key Expertise:

- **Technologies:** React.js, TypeScript, Node.js, Ruby on Rails, and SQL databases.
- **Cloud & Architecture:** Strong expertise in cloud integration (EC2, Docker, Lambdas, API Gateway, SES, SQS, RDS), API design, testing automation, and scalable micro-frontend architectures.
- **Mentorship:** Love mentoring, helping others step up their game, and providing mock interviews and tips (verified by **tutorial YouTube channel**)

Experience

Nolte.io | Staff Lead FullStack Engineer

Standardized **Sapey.AI** platform for nursing home assessments, resulting in a **40%** reduction in manual data entry errors and ensuring compliance with regulatory reporting standards (e.g., MDS/RAI).

- Engineered a scalable, micro-frontend architecture using React and TypeScript to integrate disparate assessment modules (e.g., Fall Risk, Nutrition, Cognitive Status) into a single, cohesive user experience.
- Implemented real-time data synchronization between the frontend and a Ruby on Rails backend, reducing assessment completion time by **25%** and providing instant access to patient risk profiles for clinical staff.
- Developed a comprehensive REST API to facilitate seamless integration with existing Electronic Health Record (EHR) systems.
- *Skills Used:* React.js, TypeScript, Ruby on Rails, PostgreSQL, REST API, Micro-frontends, Datadog, Jest.

Full-stack Web Developer

LifeStance Health | Dec 2021 – Dec 2025 (48 months)

- Improved communication and optimized Jira tickets, reducing UI development time by approximately **30-40%** through precise designs and clear development workflows (Figma, React).
- Simplified codebase by up to **~30%** via React libraries code-refactoring (e.g., React-router-dom v6), making maintenance easier and reducing bugs.
- Reduced runtime errors by **~35-50%** through Javascript to Typescript code refactoring and static type checking, improving application reliability.
- Built interfaces for more than **5 new OBIE modules**, including Credit Card On File functionality.
- Reduced error rates in third-party API responses by **~40%**, enhancing frontend performance and reliability (using BugSnag and DataDog).
- Maximized patient intake-to-booking conversion rate by **~31%** through optimized workflows and seamless integration of frontend and backend systems.
- Reduced boilerplate code by approximately **~40-50%**, simplifying state management and improving developer productivity (implementing Redux Toolkit).

- *Skills Used:* Redux.js, TypeScript, React.js, Ruby on Rails, gRPC, Cypress, Unit Testing, Datadog, Jest, Express, Docker, AWS, Lambdas, CI/CD Pipelines.

Full-stack Web / Mobile Developer

AlejandroDeLaRocha.com | May 2013 – Present (148 months)

- Developed "[GetCertified](#)" iOS/Android App for cloud certification mock exams.
- Built LLM Prompt to JSON consumable APIs and pipelines using Gemini models with Gemini CLI.
- Teacher education SaaS platform [PlaneaSec.com](#) (Since 2025)
- Alexium AI Bot integrated in multiple sites attachable to databases (read only)
 - [JerrysAcuarios.com](#)
 - <https://alejandrodelarocha.com/clientes/repysol/> (Bot integrated)
- [KevinPicks.com](#) - Algorithmic AI sports odds provider
- [NegasHosting.com/edit](#) - Co-founder web hosting with it's own web builder (from scratch)
- [AltFiConstructora.com](#) - Backoffice ERP and responsive web design with editable gallery.
- Launched "Pinchimundo" Mobile app (Podcast player, iOS/Android) with over **10,000** downloads.
- *Skills Used:* React.js, MySQL, Bootstrap, Ionic Framework, Javascript, PHP, CSS3, Wordpress, Gemini AI. EC2, Docker, Lambdas, API Gateway, SES, SQS, RDS

Full-stack Web Developer

Health Services of Chihuahua State | May 2020 – May 2022 (24 months)

- Wrote inventory/pharmacy system used by **21** hospitals (including medication dispenser, inventory management, and CrystalReports).
- Automated surveys and studies, reducing data collection time by over **60%** compared to traditional methods.
- Reduced public pharmacy line waiting times for over **20%** of patients by digitalizing inventory, enabling real-time stock updates, and streamlining prescription fulfillment.
- Initiated React.js migration for existing systems.
- *Skills Used:* React.js, Javascript, Bootstrap, jQuery – React transition, Redux, Oracle Database, .NET Framework. EC2, Docker, Lambdas, API Gateway, SES, SQS, RDS

Graphics and IT Planning Supervisor

Servicio Nacional de Empleo (National Employment Service) | May 2020 – May 2022 (24 months)

- Managed network nodes mapping (database and graphic map).
- Designed **7 bimestral** magazines, significantly increasing the reach of job opportunities to **8,000** readers by leveraging key graphic design elements.
- *Skills Used:* PHP, MySQL, Adobe Photoshop.

Education

Universidad Tecnológica de Chihuahua

Engineer, Information and Communication Technologies – 2019

Focus: Software Engineering, Databases, SOLID, DRY, and Domain Driven Design patterns.

Technical Skills

Category	Skills
AI Integration	Claude API, Large Language Model (LLM) prompt engineering, Retrieval-Augmented Generation (RAG), vector databases (Pinecone, Weaviate), embeddings, semantic search, LangChain, streaming response implementation, function calling, AI agent development, bespoke chatbot solutions, speech-to-text and text-to-speech technologies (Whisper, ElevenLabs), Twilio Voice, WebRTC, real-time AI interface design, REST API and webhook integration, Node.js AI backend development, serverless AI functions, rate limiting, token optimization, AI cost optimization, model fine-tuning workflows, model evaluation, and AI workflow automation.
Programming Languages	JavaScript, TypeScript, SQL, Ruby, SASS, PHP
Frontend & User Interface (UI)	React.js, Redux.js, Next.js, Vue, AngularJS, Ionic Framework, Bootstrap, Tailwind CSS, Vite, Webpack, Babel.js, Figma
Backend & APIs	Node.js, Express.js, Ruby on Rails, PHP, .NET Framework, RESTful architectures, HTTP/2, Apollo GraphQL, gRPC
Cloud & Deployment	Amazon Web Services (AWS) including EC2, S3, and Lambda functions, Docker containerization, Relational Database Service (RDS), and Continuous Integration/Continuous Deployment (CI/CD) with GitHub pipelines.
Databases	MySQL, PostgreSQL, Oracle SQL Server
Testing & Quality Assurance (QA)	Cypress, comprehensive Unit Testing methodologies, Jest, Datadog
Version Control & Package Management	Git, Yarn, Pnpm, npm